

RSNA Knee MRI: report-supervised abnormality detection

Sprint 03 project report - Deep Learning, NLP and Transformers (Nolyth) | Abdul Salam | October 2026

Summary. The task is to predict 12 knee findings from MRI studies (Kaggle: RSNA Knee Abnormality Detection). Only 58 of 4,407 training studies have real labels, so radiology reports in nine or more languages were used to create training labels. We compared a rule-based labeler, TF-IDF with logistic regression and a multilingual DistilBERT, then trained image models on the resulting soft labels. The final ResNet18 with attention pooling reached a macro AUC of 0.692 on the 58 gold studies (95% interval 0.642 to 0.742) and 0.730 on the Kaggle public leaderboard. The work is documented in five notebooks and a deployed Streamlit demo with user accounts.

1. Problem and data

Each knee MRI study must be scored for 12 findings: ACL and MCL injury, medial and lateral meniscus tear, medial, lateral and patellofemoral osteoarthritis, effusion, synovitis, Baker's cyst, bone contusion and fracture. The metric is the macro-average of the ROC AUC over the 12 findings. Training studies come with a free-text radiology report; test studies (about 1,300) contain images only, so the final model must be image-only.

Data fact	Value
Training studies / DICOM series	4,407 / 24,371 (about 5.5 series per study; sagittal, coronal and axial planes)
Studies with real labels (gold)	58 (all 12 labels). The other 4,349 studies have only a report.
Report languages (automatic detection)	en 1,736; es 682; tr 546; hr 406; el 321; de 262; bg 220; nl 153; fr 81
DICOM format	All 6,000 sampled series use Explicit VR Little Endian; images from 256 to 1024 pixels

Core idea. Use the reports to create training labels for thousands of unlabeled studies, then train an image-only model on those labels and evaluate it honestly on the 58 studies that have real labels. These 58 studies were never used for training or model selection.

2. Approach and workflow

Stage	Notebook	What it does
Audit and rule labeler	01	Data audit, rule-based labeler v1 and v2, evaluation on gold
Text models	02	Rules vs TF-IDF vs multilingual DistilBERT; cross-fitted soft labels
Preprocessing	03	Series selection and slice cache (7.8 GB, 0 failures)
Image models	04	CNN vs ResNet18, mean vs attention pooling, bootstrap intervals
Offline inference	05	Kaggle code submission with Internet off
Demo	demo/	Streamlit app with accounts, history and Neon PostgreSQL

3. Report labeler (NLP)

3.1 Rule-based labeler

Keyword patterns for English, Spanish, German and Turkish are applied clause by clause, with negation handling ("no fracture", "ACL intact"). Version 1 scored a mean F1 of 0.536 on the 58 gold studies. The error analysis showed four problems: reports in uncovered languages received all-zero labels (27% of studies), Greek reports use a lookalike micro sign that broke the patterns, osteoarthritis is usually written as cartilage loss in a compartment rather than the word "osteoarthritis", and grade 1 sprains were counted as tears. Version 2 added Unicode normalisation, tear-only rules for ligaments and menisci, cartilage and compartment rules with context carried across clauses, and marks uncovered languages as unlabeled instead of negative. It reached a mean F1 of 0.696 on the 46 gold studies in covered languages (half A 0.723, half B 0.652, used as an overfitting check).

3.2 TF-IDF and the multilingual transformer

TF-IDF with character n-grams (3 to 5) and logistic regression was the classical baseline. The transformer was **distilbert-base-multilingual-cased**, an **encoder-only** model, a smaller distilled version of multilingual BERT pretrained on about 100 languages. Text is tokenised with its shared WordPiece vocabulary (maximum 384 tokens). Self-attention lets every token weigh every other token, which gives contextual representations. For this task a classification head with 12 sigmoid outputs was added and the whole model was fine-tuned with a weighted binary cross-entropy (8 epochs, learning rate 6e-5, mixed precision). The first attempt with 3 epochs agreed with the rules at only F1 0.48 and was not used. After retraining, agreement was 0.794. All text models were trained on rule-derived labels and scored on the same gold studies.

Method (gold set)	F1 all 58	F1 covered (n=46)	F1 other languages (n=12)	Macro AUC
Rules v2	0.587	0.695	0.000	0.698
TF-IDF + LR	0.547	0.629	0.066	0.713
DistilBERT multilingual	0.571	0.616	0.299	0.713

The rules are best on the covered languages, because the other methods are trained on rule labels and cannot beat their teacher there. DistilBERT is the only method that scores on the 12 studies in other languages (F1 0.299, from a tiny sample). A blend of rules and transformer reached a macro AUC of 0.743. For training, labels were produced with 3-fold cross-fitting (each study predicted by a model that never saw it): soft label = 0.5 x rule + 0.5 x transformer for covered languages, and the transformer alone for the others. This gave 4,349 training studies.

4. Image model

4.1 Preprocessing

For each study one fluid-sensitive series per plane was chosen, the slices were sorted by their position in the DICOM header, 16 evenly spaced slices from the central 60% were kept, intensities were clipped at the 1st and 99th percentile per series, and images were resized to 192x192 and stored as uint8 (3 planes x 16 slices, 7.8 GB for all studies, no failures).

4.2 Models and training

A shared 2D backbone encodes every slice; the slices of a plane are pooled (mean or attention); the three plane vectors are concatenated and a linear layer outputs 12 scores. Three models were compared: a small CNN trained from scratch, ResNet18 with ImageNet weights adapted to one channel (transfer learning) with mean pooling, and the same ResNet18 with **attention pooling**, which learns a weight per slice. Training used 3,915 studies, 434 for validation, AdamW with a one-cycle schedule, mixed precision, a weighted loss for rare findings, brightness and contrast jitter and small shifts. Flips were not used because they would swap medial and lateral. The best epoch was chosen by validation loss.

Image model	Gold macro AUC	95% bootstrap interval	Validation AUC vs labels
Small CNN from scratch	0.631	0.568 - 0.690	0.725
ResNet18, mean pooling	0.672	0.614 - 0.725	0.813
ResNet18, attention pooling (final)	0.692	0.642 - 0.742	0.796

Transfer learning clearly beat training from scratch. Attention and mean pooling are statistically indistinguishable: attention is better on gold and mean is better on validation, and the intervals overlap. Averaging two models did not help (0.685), so the single attention model was submitted. The offline Kaggle submission, run on the hidden test set with Internet disabled, scored **0.730 on the public leaderboard** (about 30% of the test data), inside the gold interval.

5. Per-finding results

Finding	Image AUC (final)	Rule F1	Finding	Image AUC (final)	Rule F1
ACL	0.69	0.92	PF OA	0.66	0.43
MCL	0.63	0.67	Effusion	0.70	0.73
Medial meniscus	0.60	0.85	Synovitis	0.61	0.61
Lateral meniscus	0.71	0.89	Baker's cyst	0.78	0.71
Medial OA	0.76	0.67	Contusion	0.67	0.63
Lateral OA	0.85	0.63	Fracture	0.63	0.61

Values come from 58 gold studies, so rare-finding scores are rough. Lateral OA, Baker's cyst and medial OA are the strongest image results; medial meniscus, synovitis, MCL and fracture are weakest.

6. Error analysis

Labeler

- Remaining effusion false positives: in the four cases inspected, the reports describe small or moderate effusions that the gold labels count as negative.
- Gold labels can contradict the report text: one Spanish report states a tear of the medial and lateral meniscus, while the gold medial meniscus label is 0. Reports are therefore a noisy proxy and F1 near 1.0 is not reachable.
- Patellofemoral osteoarthritis is the weakest rule finding (F1 0.43) because cartilage defects of the patella are described in many different ways.

Image model

- In a worked example (gold study 29), 6 of 12 verdicts at the 0.5 threshold were wrong, and all 6 were false positives. No true finding was missed.
- The scores cluster in a narrow band (mostly 0.45 to 0.67). The model ranks findings only moderately well, which matches an AUC of about 0.69.
- The weighted loss raises probabilities for rare findings, so a fixed 0.5 threshold over-predicts. Outputs are relative scores, not probabilities of disease.
- The checkpoint selected by validation loss was an early epoch, while validation AUC kept improving to epoch 6; selecting by AUC may help.

7. Deployed demo

A Streamlit application (frontend and application server) with a Neon PostgreSQL database (SQLite in development) lets users sign up and log in, paste a radiology report to see the detected findings with the triggering sentence, upload a zipped DICOM study to run the final model, and review their own test history. An admin page lists users and their tests. Passwords are stored as salted PBKDF2 hashes, repeated failed logins lock the account for 10 minutes, and uploaded files are not stored. The demo reports the measured reliability of each finding next to its score and states that outputs are not a diagnosis. Automated tests cover authentication, per-user history, the report labeler and the DICOM pipeline. The fine-tuned DistilBERT was not saved, so the demo's report tab uses the rule labeler.

8. Limitations

- Only 58 gold studies: differences such as 0.672 versus 0.692 are within the noise, and per-finding values are rough.
- Training labels come from reports and inherit their errors; the rule labeler covers four languages.
- The public leaderboard score covers about 30% of the test data and may change on the final data.
- The probability threshold was not tuned and outputs should be read as rankings.
- The model is for research and learning only and must not be used for clinical decisions.

9. What was learned and next steps

- Weak supervision: when labels are scarce, a validated labeler plus honest evaluation on a small gold set is a workable strategy.
- Careful error analysis improved the labeler more than model changes (F1 0.536 to 0.696).
- Transfer learning and a pretrained multilingual transformer both helped; attention pooling was a useful, if statistically tied, extension.
- Next steps: select checkpoints by AUC, calibrate per-finding thresholds on validation data, improve cartilage and effusion rules, save the fine-tuned transformer for the demo, and try more slices per series.

Reproducibility

Code: github.com/salamlakhan7/rsna-knee-mri-multimodal (notebooks 01 to 05, demo, results).

Kaggle datasets hold the pseudo-labels, soft labels, slice cache and model weights. Competition data is not redistributed in the repository.